RESEARCH / PROJECT ETHICS REVIEW FORM
For Thesis / Capstone / Dissertation Proposals
(*ver.Oct2024c*)

Names of Student Researcher(s):

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Research/Thesis/Capstone Title: A New Rendering and Plume Simulation Framework for Visualizing Taal Volcano Eruption Plumes

College: College of Computer Studies

Department: Software Technology

Course: BS Computer Science major in Software Technology

Expected Duration of the Project: from: Aug 6, 2026 to Aug 6, 2027

PROTOCOL:

(*To be filled up by proponents; Use the Ethics Checklists as guides in determining areas for ethical concern/consideration. Attach and properly reference the accomplished ethics checklists and supporting documents.)*

		Reference Document (include page number)
Target participants	**Sample Size**: 10 to 20 voluntary participants, recruited according to a tiered priority order: PHIVOLCS personnel as the primary target group, followed by individuals who witnessed a Taal Volcano eruption firsthand, then volcano/geohazard simulation enthusiasts, and lastly computer science students with a background in rendering or volcano-related coursework, should earlier priority groups be insufficient to reach the target sample size. **Inclusion Criteria**: Recruitment will follow a tiered priority order. Priority 1: PHIVOLCS personnel with domain expertise in volcanology and hazard assessment. Priority 2 (if Priority 1 pool is insufficient): individuals who witnessed a Taal Volcano eruption firsthand. Priority 3: enthusiasts with experience using volcano/geohazard simulation software. Priority 4: computer science students or developers with a relevant	Scope and Limitations (page 4-5)

	background in 3D graphics, rendering, or volcano-related coursework/projects. **Exclusion Criteria**: Individuals who do not fall into any of the four priority categories above, or who are unable to complete the informed consent process. **Withdrawal Criteria**: Participants may exit the software at any stage and can choose to withdraw from the evaluation without academic or personal penalty.	
Data to be used / collected	**Technical Data:** Empirical hardware performance logs including Frame Rate (FPS), CPU/GPU resource utilization percentage, and frame rendering times (ms). **Human Data:** Quantitative usability scores via the System Usability Scale (SUS) questionnaire and qualitative feedback regarding visual artifacts or rendering anomalies. No personally identifiable information (PII) such as names, student numbers, or email addresses will be tied to the responses. Also, feedback gathered from a closing interview regarding suggestions for improving the interface and overall performance.	Scope and Limitations (page 4-5)
Brief procedure	1. **Screening:** Interested participants complete a short pre-screening form confirming which priority category they fall under (PHIVOLCS personnel, Taal eruption witness, volcano/geohazard simulation enthusiast, or CS student/developer with relevant graphics background), consistent with the tiered recruitment order. 2. **Consent:** Eligible participants are presented with the Informed Consent Form (ICF), which outlines the evaluation's scope, hardware interactions, session length, and data privacy measures. 3. **Orientation/Briefing:** Participants are given a brief, neutral walkthrough of	Scope and Limitations (page 4-5)

	the interactive Taal Volcano Plume simulator's controls and features. 4. **Usability Task:** Participants will use the software to recreate specific Taal Volcano eruption plumes in response to predefined scenarios. 5. **Post-Session Questionnaire :** Participants will complete an anonymous, self-administered SUS questionnaire to rate the software's ease of use and compare visual consistency. 6. **Feedback:** A closing interview will be conducted to gather suggestions for improving the interface and overall performance.	
Potential risks	**Physical**: Minimal risks, though participants may experience minor eye strain or visual fatigue from interacting with the 3D graphics simulator. **Technical**: Participants may experience frustration if they encounter bugs or usability issues with the experimental prototype. **Mitigation**: Researchers will provide immediate technical support for major difficulties, and participants are encouraged to request breaks at any time if they feel tired or frustrated. **Psychological:** For participants recruited as eruption witnesses, engaging with a simulation recreating the Taal eruption may evoke discomfort or distress related to their lived experience. Mitigation: The ICF for this participant category will explicitly disclose the nature of the simulated content in advance, participants will be reminded they may stop or withdraw at any point without penalty, and researchers will monitor for visible signs of distress during sessions.	

Applicable Terms of Use or Licensing Agreement of Data, Models and/or Tools	**AnitoPlume Source Code:** Used with explicit permission and under the supervision of the thesis adviser, Neil Del Gallego**.** **Libraries:** Eigen (Mozilla Public License v2.0) and dearIMGUI (MIT License). **Graphics Toolkits & SDKs:** Khronos Vulkan SDK, Microsoft DirectX 12 SDK, and Khronos OpenGL specifications. **Profiling Tools:** RenderDoc (MIT License).	Scope and Limitations (page 4-5)
Steps to safeguard the participants and/or data	**Participant Safety:** Testing sessions will be kept brief to prevent eye strain. Participants will be explicitly informed via the ICF that participation is entirely voluntary and that they may withdraw from the usability test at any point without academic or personal penalty. Participants recruited as PHIVOLCS personnel will be engaged through formal coordination with the agency, with a permission letter obtained and session scheduling arranged to minimize disruption to their official duties. **Data Safety:** All performance logs and usability questionnaire responses will be fully anonymized using alphanumeric identifiers. **Data Storage:** Survey sheets and digital spreadsheets will be secured in a password-protected cloud storage repository accessible only to the primary student researchers. **Data Retention:** Anonymized data will be kept 4 to 5 years following final publication or degree conferral to allow for reproducibility audits before permanent deletion	Scope and Limitations (page 4-5)
Other concerns and considerations	None at this point in time	

PANEL'S ASSESSMENT/COMMENTS:

PANEL'S DECISION:

- € **Approved** (*Proponent/s may proceed with the study/data collection*)
- € **Modifications Required** (*Proponent/s should address ethical issues, follow the panel's recommendations for required modifications, and resubmit Ethics Review Form for approval before proceeding with the study or before starting with data collection*)
- € **For Full Review** (*Proponent/s should submit the proposal to the University Research Ethics Review Committee (RERC) and obtain a research ethics clearance before proceeding with the study or before starting with data collection*)

JUSTIFICATION:
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Name and Signature of Adviser/Mentor

Date: ______________________

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Name and Signature of Panel Member	**Name and Signature of Panel Member**
Date: ______________________	**Date:** ______________________

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Name and Signature of Panel Member

Date: ______________________